

SoilSidekick Pro® / LeafEngines™ SDK

Technical Overview for AFRL/RYSM Evaluation

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Mission

The LeafEngines™ SDK delivers micro-local environmental intelligence through a unified, privacy-preserving API stack. It enables partners to integrate real-time soil, vegetation, and environmental sensing into existing systems without ML expertise, and supports fully offline operational modes for contested, denied, or bandwidth-limited environments.

Core Capabilities

Multi-Modal Environmental Intelligence

- Soil composition, moisture, compaction, and structure inference
- Vegetation health with species-level indicators and stress detection
- Micro-climate and terrain-aware contextual reasoning
- Sensor-agnostic ingestion (RGB, NIR, multispectral, geospatial context)

Unified API Surface

- Single endpoint for all inference tasks with standardized outputs
- Regulatory-grade reasoning trace (transparent, auditable outputs)
- Supports batch, streaming, and edge-device workflows

Developer-Friendly Integration

- SDKs for iOS, Android, Web, and Python — integrates in hours, not weeks
- Lightweight, serverless-optional architecture; no specialized hardware required
- SOC 2-aligned security with role-based access control

Architecture Overview

LeafEngines™ acts as the enterprise-grade API layer embedding SoilSidekick Pro® capabilities into a pipeline:

1. Input: Image(s), sensor metadata, geolocation, optional auxiliary data
2. Pre-processing: On-device or hybrid (privacy-preserving, no raw data leaves device)
3. Inference: Patent-protected multi-modal reasoning engine
4. Output: Structured environmental intelligence with confidence intervals and uncertainty bounds
5. Sync: Offline caching with delayed sync or air-gapped deployment

Operational Modes & Security

Offline-First	Full functionality without network connectivity. All inference executes on-device.
Hybrid	Local inference + cloud enhancement when available. Graceful degradation on disconnect.
Edge / Air-Gap	Low-power, low-bandwidth environments. Optional on-prem or air-gapped deployment.

Security: Privacy-preserving by design. No raw imagery leaves device. SOC 2-aligned controls. Patent-protected inference (no weights exposed). Air-gapped deployment supported.

DV005 Objective Alignment

Mapping SDK Capabilities to AFRL/RYSM DV005 Requirements

DV005 Objective	SDK Capability	IP Protection	TRL	Status
1. Geo-Specific Scene Construction	Vincenty geodesy engine computes sub-meter displacements on WGS-84 ellipsoid. AlphaEarth 64-dim semantic embeddings at 10m resolution provide ground-truth environmental priors.	CIP #19/544,827 Claim 1, S106	6	Ready
2. Sensor-Independent Multimodal Output	EO/IR and RF modalities rendered as co-registered projections of a single environmental state vector (moisture, NDVI, conductivity). Shared physical priors enforce consistency.	Parent #19/320,727 + CIP S102	5-6	Ready
3. Cross-Modal Physical Consistency	Complementary filter blends high-frequency inertial orientation with low-frequency magnetometer corrections. Real-time Pearson correlation validates cross-modal coherence.	CIP Claim 1, S102/S112	6	Ready
4. Uncertainty Propagation	Kalman-inspired variance model propagates positional uncertainty per step. Reliability gating prevents low-confidence data from persisting to storage or downstream consumers.	CIP Claim 1, S108/S110/S112	6	Ready
5. DIL-Resilient Offline Execution	100% deterministic math executes on-device with zero network dependencies. Exportable as standalone HTML. Three-tier fallback chain.	CIP Claim 1, S114 (Gating)	6	Ready
6. Distributed RF Sensing Support	Geodesic triangulation via Vincenty engine supports multi-node RF geolocation. Per-node uncertainty envelopes enable weighted fusion for distributed imaging arrays.	Parent #19/320,727 (Grid) + CIP S106/S108	4-5	Planned

Response: RYSM-Relevant Sensing Modalities

Per AFRL SME request, the following modalities and outputs are directly applicable to RYSM priorities:

Modality	SDK Implementation	RYSM Application
EO/IR Co-Registration	AlphaEarth 64-dimensional semantic embeddings encode land meaning (cover, moisture, vegetation health) at 10m ² resolution from multi-temporal Sentinel-2 imagery.	Synthetic scene generation with physics-grounded ground truth
RF Geolocation	Vincenty geodesic engine provides sub-meter triangulation for multi-node positioning. Per-node uncertainty envelopes support weighted fusion.	Distributed RF imaging array coordination
Uncertainty-Gated Outputs	Kalman-inspired variance propagation with write-inhibition at 500m uncertainty radius. All outputs carry confidence intervals.	Reliability-aware synthetic training data for ML pipelines

Proposed Next Steps

- 6. AFRL SMEs provide mission alignment feedback on prioritized modalities

7. SoilSidekick Pro delivers interactive offline demo (standalone HTML) for independent RYMS evaluation
8. Technical onboarding session: API walkthrough, sample integration, and field test design
9. Define minimum viable experiment (MVE): 30-min GPS-denied field test comparing DR-estimated vs. ground-truth endpoints

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